NGUYEN HUY HOANG

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Artificial Intelligence graduate with hands-on internship experience supporting computer vision model development and deployment workflows. Comfortable working independently across data preparation, model training, evaluation, and API-based inference. Experienced in documenting experiments and maintaining reproducible pipelines. Seeking a remote junior AI/ML role focused on practical machine learning systems.

TECHNICAL SKILLS

- **Programming & Data:** Python, SQL, NumPy, Pandas
- **Machine Learning & Deep Learning:** PyTorch, scikit-learn, OpenCV, object detection, image classification, model evaluation (F1, mAP, BLEU)
- **Vision-Language & NLP:** Hugging Face Transformers, Unsloth, fine-tuning Vision-Language models (Qwen2-VL, InternVL, MiniCPM), RAG fundamentals, prompt experimentation
- **Deployment & Engineering Tools:** FastAPI, Docker, Git, GitHub
- **Workflow & Collaboration:** Experiment documentation, version control best practices, comfortable with asynchronous communication

WORKING EXPERIENCE

AI Intern – CEH IT Services Co., Ltd. — Jun 2024 – Dec 2024

Ho Chi Minh City, Vietnam

- Designed and evaluated hazard detection computer vision models using PyTorch & OpenCV, actively enhancing automated safety monitoring workflows.
- Trained and optimized object detection and document-related vision models; managed dataset preparation, annotation pipelines, and rigorous performance evaluation (mAP, F1-Score).
- Processed diverse operational datasets (image/video) via rigorous data cleaning and label validation, ensuring structured storage for reproducible experiments.
- Engineered API-based deployment pipelines to integrate model inference outputs directly into enterprise systems, collaborating closely with cross-functional software teams.

PROJECTS

Vision-Language AI for Chest X-ray Caption Automation — July 2024 – Dec 2025

PyTorch, Hugging Face, Qwen2-VL, BLEU

- Developed an assistive Vision-Language system to automatically generate structured Vietnamese medical reports (Findings and Impressions) from chest X-rays.
- Fine-tuned foundational VLMs (Qwen2-VL, InternVL) on a proprietary, anonymized hospital dataset of 137,930 examinations.
- Set up a robust inference pipeline via FastAPI and NVIDIA Triton, achieving **89.2%** BLEU score and **74.5%** blind-pass rate on clinical **"Doctor Benchmarks"**.

CARLA SAC Autonomous Driving — Jun 2025 - Sep 2025

Python, Stable Baselines3, Optuna, Reinforcement Learning

- Trained a Soft Actor-Critic (SAC) reinforcement learning agent within the CARLA Simulator for complex end-to-end driving tasks.
- Implemented Curriculum Learning and compressed 50FPS raw visual observation streams into a 95-dimensional continuous latent space using a customized FP16 Variational Autoencoder (VAE).
- Automated hyperparameter tuning via Optuna, ultimately achieving an **85%** route completion success rate while navigating dynamic NPC traffic networks.

CERTIFICATIONS

- NVIDIA Deep Learning Institute: Building Real-Time Video AI Applications | Accelerating End-to-End Data Science Workflows
- Udemy: Machine Learning & Deep Learning (Python/R) | Python AI: Smart Assistants & Computer Vision
- Coursera: IBM AI Enterprise Workflow

EDUCATION

FPT University

2021 - 2025

Bachelor Degree of Artificial Intelligence and Data Science at FPT University

- GPA: 3.77/4.0
- Honors: Academic Honors (Spring 2025, Summer 2024)
- Awards: Global Hackathon Participant (AngelHack Virtual Badge)
- Graduation Project: Vision-Language AI for Chest X-ray Caption Automation